Smart Hostel Maintenance Management System: An Intelligent Scheduling and Request Management Platform

Problem Statement

The current hostel maintenance process at IIIT-Delhi primarily relies on WhatsApp communication with the Building Management Services (BMS) team. Students submit requests such as room cleaning, deep cleaning, painting, electrical repairs, plumbing issues, AC servicing, key collection, cupboard installation, and other maintenance tasks. Typically, students receive only a **"Noted"** response, with no information regarding the expected service time, request status, or whether maintenance staff have already visited.

Since students spend a significant part of the day attending classes, labs, and other academic activities, their rooms are often locked during maintenance visits. As a result, staff frequently make unsuccessful visits, while students remain unaware that the staff had already come. This leads to repeated follow-ups, delayed service, inefficient utilization of maintenance staff, and poor communication between students and BMS.

Problem 1 : endless loop of noted

Problem 2 : mail/message different departments for similar needs

Problem 3 : wastage of resources/time and inconvenience for students and cleaning staff

Problem 4 : no priority for guest

Proposed Solution

Develop a **Smart Hostel Maintenance Management System**, a web/mobile platform that replaces the current unstructured WhatsApp-based workflow with a structured maintenance management system.

The platform will allow students to:

- Submit maintenance requests by selecting the appropriate service category (room cleaning, deep cleaning, painting, plumbing, electrical, AC repair, key collection, cupboard installation, etc.).
- Select a preferred service slot (Morning, Evening , Night).
- Receive an allocated service window based on staff availability.
- Track the status of their request in real time (Received → Assigned → In Progress → Completed).
- Reschedule requests when they are unavailable.

Maintenance staff will receive a prioritized task list, while administrators will be able to monitor requests, assign staff, and analyze system performance through a centralized dashboard.

Intelligent Scheduling Framework

The core research contribution of this project is the **maintenance scheduling engine**.

Each maintenance request will be modeled as a scheduling problem similar to process scheduling in Operating Systems. Different scheduling algorithms such as **First Come First Serve (FCFS), Round Robin, Priority Scheduling**, and custom heuristic-based approaches will be explored to determine the most efficient method for assigning maintenance requests.

The scheduler will consider multiple factors, including:

- Request priority
- Staff availability
- Student-preferred time slots
- Hostel block and floor
- Current staff workload

The system will also support **dynamic priority assignment**. For example:

- Emergency requests such as electrical faults, water leakage, or non-functional fans/ACs during summer will receive higher priority.
- Requests from guest house occupants or visiting faculty can be assigned elevated priority when required.
- Routine services such as regular room cleaning will be scheduled while maintaining fairness among students.

Research Objectives

- Design an intelligent scheduling framework for hostel maintenance requests.
- Compare different scheduling algorithms to optimize request allocation.
- Reduce unsuccessful maintenance visits caused by locked rooms.
- Improve communication and transparency between students and maintenance staff.
- Evaluate the effectiveness of the proposed scheduling strategies using real-world performance metrics.

Evaluation Metrics

The proposed system will be evaluated using:

- Average request completion time.
- Number of unsuccessful maintenance visits.
- Staff workload distribution and utilization.
- Student waiting time.
- Student satisfaction through feedback surveys.

Expected Outcomes

- Reduced failed maintenance visits.
- Improved communication and transparency throughout the maintenance process.
- Better utilization and scheduling of maintenance staff.
- Faster resolution of high-priority maintenance requests.
- Data-driven insights for BMS through maintenance analytics.
- A scalable framework that can be extended to other campus service requests such as IT support, housekeeping, and facility management.

Future Scope

The proposed framework can be extended beyond hostel maintenance to support other campus services requiring task scheduling and resource allocation. It can also incorporate predictive scheduling using historical maintenance data to further improve operational efficiency.

Semester 1 — Development & Initial Framework

Goal: Develop the core Smart Hostel Maintenance Management System and establish the initial scheduling framework.

- Design and develop the **web-based platform** for students, maintenance staff and administrators.
- Implement request submission, categorization, preferred slots, status tracking and rescheduling.
- Develop the backend, database and maintenance-request management system.
- Develop the initial scheduling engine.
- Implement and study **FCFS, Round Robin and Priority Scheduling**.
- Set up the simulation/data framework and define evaluation metrics.

End of Semester 1:
A **functional end-to-end system with an integrated scheduling framework**, along with baseline algorithms and initial experimental results.

Semester 2 — Intelligent Scheduling & Evaluation

Goal: Improve the scheduling framework and evaluate its effectiveness using realistic maintenance scenarios/data.

- Develop the proposed/custom scheduling approach.
- Incorporate constraints such as **priority, staff availability, workload, location and preferred time slots**.
- Compare the proposed approach against the baseline algorithms.
- Evaluate waiting time, completion time, unsuccessful visits, staff utilization, workload distribution and student satisfaction.
- Refine the system based on experimental findings.

End of Semester 2:
A **complete Smart Hostel Maintenance Management System with an evaluated intelligent scheduling approach**, supported by comparative experimental results and analysis.